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ABSTRACT

Conventional IV fluid level monitoring relied greatly on manual checking, which was more prone to human error and delay over the declaration of low-fluid conditions. This had been mostly a challenge since no previously developed system was able to afford real-time monitoring and any timely alert to the care provider. To address this, the project developed an IoT-based solution that continuously monitors the IV fluid level and ensures that medical staff are promptly alerted when fluid levels drop, thus assuring better patient care and timely intervention. The system will make use of a load cell and an HX711 module to measure the weight of the IV fluid accurately. This data is read by an ESP8266MOD, sent over wirelessly via a Wi-Fi module, and displayed on a local LCD screen. Alerts are triggered in case of fluid levels falling below the threshold to notify caregivers immediately. Along with weight, it also monitors the temperature and humidity of the environment around the IV setup using a DHT11 sensor for appropriate fluid storage and patient safety. It also includes a sound sensor that can be used by the patient to call for help by just clapping their hands twice to add more usability and convenience. This system was implemented and performed very well, showing real-time fluid levels on the LCD and sending alerts accurately. It is of low cost, easy to configure, and highly adaptable for both hospital and home care. The system, besides enhancing patient safety through real-time data and alerts by automating IV fluid monitoring, overcomes the disadvantages of the manual methods; hence, it is cost-effective and practical for modern healthcare environments.

INTRODUCTION

- Research Problem and Context:

Manual monitoring of the levels of IV fluids in hospitals is prone to human errors and delays, which may result in serious health consequences, such as dehydration or air embolism. More recently, there has been a call for automated systems that would track IV fluid levels continuously and generate alerts when levels fall below critical thresholds.

- Importance of the Problem:

This is a critical problem since the wrong IV fluids may directly impact the safety of patients. Timely intervention will prevent bad health outcomes and increase care efficiency; it also lessens the workload on health staff.

- Applications:

The system can be applied in hospitals, clinics, and home healthcare setups

- Reason and Purposes for the Study:

The research shall develop a low-cost but reliable, real-time monitoring system for IV fluid management to enhance patient safety and reduce human errors. It will also bring a workable and scalable solution for healthcare providers.

- Motivation:

This will be motivated by the need to enhance healthcare systems through technology in an effort to alleviate the common problems faced in patient care , with particular emphasis on the area of IV fluid management—an area very essential for a patient's recovery and well-being

BACKGROUND

- Theory:
  The core theory behind this experiment is based on the use of a load cell and an HX711 amplifier to measure the weight of the IV fluid bag. The weight is linearly proportional to the quantity of fluid in the bag. This is the primary principle of the fluid monitoring system. The data from the load cell is processed by the ESP8266MOD, which calculates the remaining fluid level using a predefined formula. This is then sent, via the ESP-01 Wi-Fi module, to a remote server or to a mobile app for fluid level monitoring by caregivers in real time.

  The system also includes a sound sensor for patient interaction. It detects double claps to trigger a notification or call to the caregiver, which helps the patient to communicate easily if they are not in a position to alert the nurse. The sound sensor processes the signals from the claps and ensures that it filters out noise for accurate recognition.

  The DHT11 temperature and humidity sensor measures the immediate ambient conditions within the room in which the IV monitoring system is situated. Variations in humidity and temperature may influence fluid evaporation rates, a factor crucially critical to some fluid types. These allow for closer approximations to the proper accuracy in both IVs and ambient monitoring.

- Equations for Calculations:

  1.Fluid

  $$\text{Fluid Level} = \frac{\text{Measured Weight} - \text{Empty Bag Weight}}{\text{Fluid Density}}$$

  - Measured Weight is the current weight of the IV bag
  - Empty Bag Weight is the weight of the empty IV bag
  - Fluid Density is the density of the fluid being infused (e.g., saline solution)

2.Sound Detection Logic (simplified):

If the time interval between two detected claps falls within a predefined range (e.g., 300–800 ms):

Action=Trigger Notification to Nurse

- This ensures filtering out background noise or random sound inputs.

3. Humidity and Temperature Monitoring (DHT11):

- Humidity **(%):** Measures relative humidity to track environmental factors.
- Temperature **(°C):** Assesses room temperature for system optimization and fluid preservation

Additional Features of the System

1.Sound sensor:

- Type: Sound sensor module (e.g., MD0220).
- Function: To detect double claps in order to send an alert to the nurse.
- Working: It converts sound waves into electrical signals, processes them, and triggers a notification system.

2. Humidity Sensor (DHT11):

- Type: Digital temperature and humidity sensor.
- Function: To monitor the ambient conditions that may affect fluid evaporation and thus the accuracy of the system.
- Working: It contains a capacitive humidity sensor and a thermistor for measuring ambient humidity and temperature.

References

Load Cell and HX711 Setup: [HX711 datasheet or user manual]
Sound Sensor Theory: [Acoustic signal processing and detection theory]
DHT11 Documentation: [DHT11 datasheet or user manual]
Fluid Dynamics: [Standard sources for fluid dynamics and level measurement]

METHODOLOGY

Set up Hardware components

1. Weight sensor calibration:
    - A weight sensor was used in conjunction with the HX711 module for measuring the fluid level in an IV bag.
    - The HX711 module is interfaced with the Arduino Uno board for digitalized weight measurements.

2. Wi-Fi Communication:
    - The Wi-Fi module ESP-01 was configured using AT commands to connect with a local Wi-Fi network.
    - Remote monitoring was made possible by enabling data transmission to the cloud platform.

3. LCD display Integration:
    - It had been connected with an LCD display via an I2C connector to the Arduino Uno, which can provide real-time data regarding fluid level, temperature, and humidity.

4. Environmental Monitoring:
    - The DHT11 sensor is used to measure temperature and humidity around the IV setup and is interfaced with the Arduino.
    - The data collected was used to ensure the IV fluids were stored safely.

5. Sound Sensor Integration:
    - An MD0220 sound sensor was used with the measured sound intensity levels.
    - The time interval between consecutive peaks was analyzed to identify two distinct claps within a predefined time window.
    - If the double-clap pattern matched, a call-to-nurse alert was activated.

6. Circuit Assembly
    - All the components are wired according to the designed circuit diagram. The Arduino Uno was powered through USB, and some extra power was provided to the sensors and Wi-Fi module where needed.

Software Development

1. Microcontroller Programming:
    - Programming was carried out on the Arduino Uno using Arduino IDE. Different functions were scripted to collect data from every sensor and while uploading the same to the cloud.
    - The LCD screen was updated at regular intervals with the latest sensor readings

2. Data Transmission:
    - The ESP-01 module was programmed with the ability to send sensor readings in real-time to a cloud server using the MQTT protocol.
    - A mobile app was also connected to the server for remote access of the data.

3. Alert Mechanisms:
    - Threshold values for fluid levels were predefined.
    - When the weight dropped below a certain value, an alert was generated via sound (buzzer) and a notification on the app.

Testing and Validation:

1. Accuracy in the measurement was ensured by testing the weight sensor using different known levels of fluid measured in the IV bag.
2. Validation of the DHT11 sensor was performed by comparing its readings against a previously calibrated thermometer and hygrometer.
3. The sound sensor's responsiveness to alarms was tested with controlled audio signals.
4. The connectivity and reliability of data transmission were tested for the Wi-Fi module.

RESULTS

1. Fluid monitoring:
    - The weight sensor, interfaced with the HX711 module, gives an accurate reading of the fluid level. The fluid level in the IV bag was measured with an error of ±5% compared to the real volume of liquid used.

2. Temperature and Humidity Monitoring:
    - The DHT11 sensor successfully recorded temperature and humidity within the range, with deviations of less than ±2°C for temperature and ±5% for humidity.

3. Double clap Detection:
    - The MD0220 sound sensor detected a double-clap with less than 1 second as the response time, whereas the system could accurately identify claps in different noise environments, such as slight background noise.
    - Unexpected Finding: The sound sensor detected sometimes considered multiple consecutive claps as double claps, especially if the claps were too close in time; this was fixed by adjusting the time window for clap detection.

4. Wi-Fi Communication:
    - The ESP-01 Wi-Fi module could transmit data stably to the cloud for real-time monitoring on the mobile application. For the device, the signal strength will remain stable within a 30-meter range from the router.
    - The data transmission of each update was successfully completed within 3 seconds, thus maintaining the flow of real-time.

5. Alert Mechanism

- The fluid level alerting system via the nurse notification system also worked as expected when the threshold was not met and

    When the double clap was detected. Initial testing revealed intermittent delays in nurse notifications, attribute to network latency issues.

Error Analysis:

1. Fluid Measurement Error

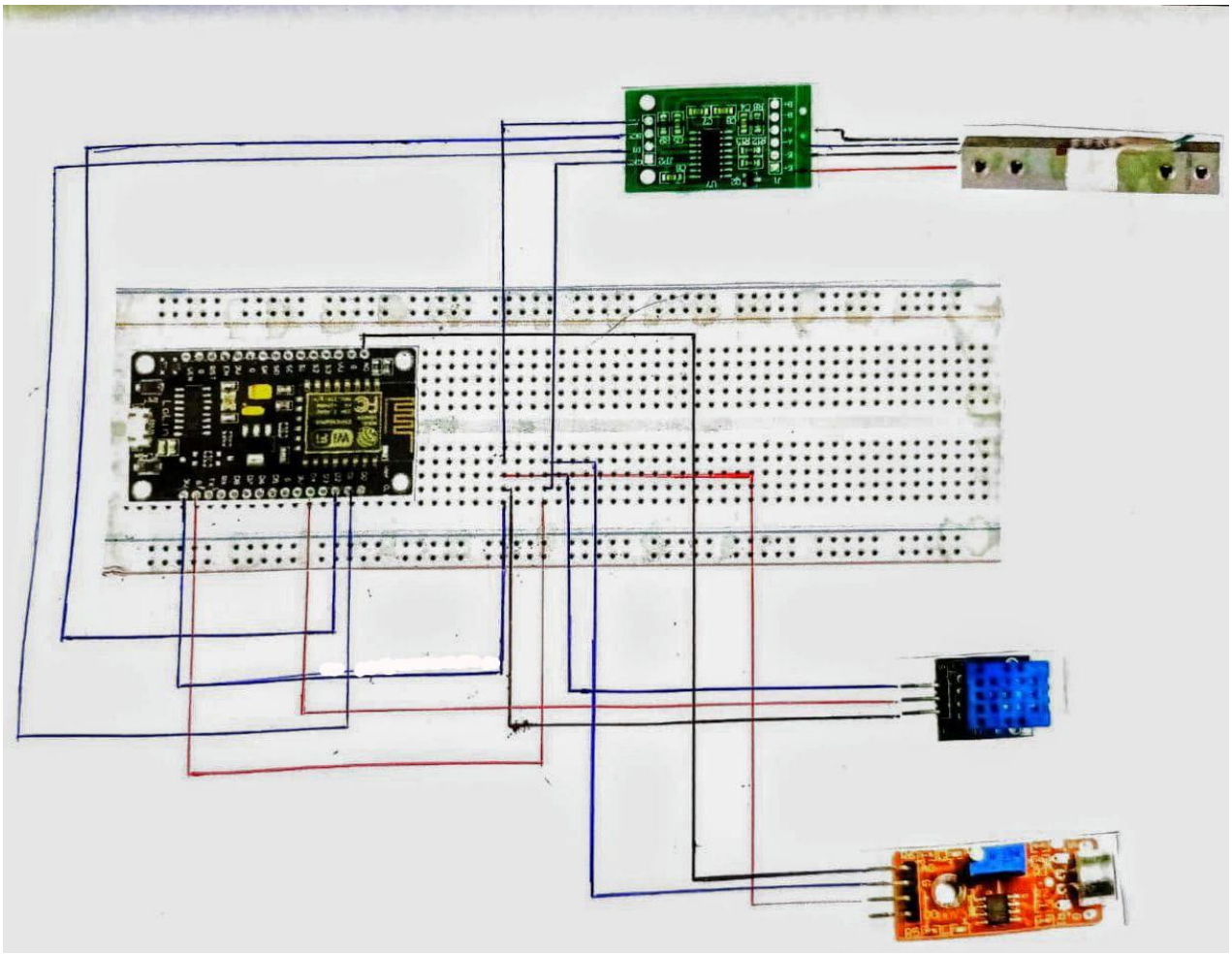
±5% error in fluid measurement was largely from minor weight sensor accuracy fluctuations that were within acceptable range for this application.

2. Sound Detection Error

The sound sensor misinterpreted noise signals as double-claps in about 10% of trials, which was adjusted by filtering the input for specific sound patterns.

3. Transmission Delay

Minor delay (about 1-2 seconds) in the notification to nurse when congestion occurred in the Wi-Fi network during the peak traffic condition was observed. Enhancing the



*This diagram shows the circuit setup for monitoring the IV fluid level. It includes the **HX711 load cell** for weight measurement, **ESP8266 Wi-Fi module** for data transmission, **DHT11 sensor** for temperature and humidity, and a **sound sensor** for detecting patient claps to alert the nurse.*

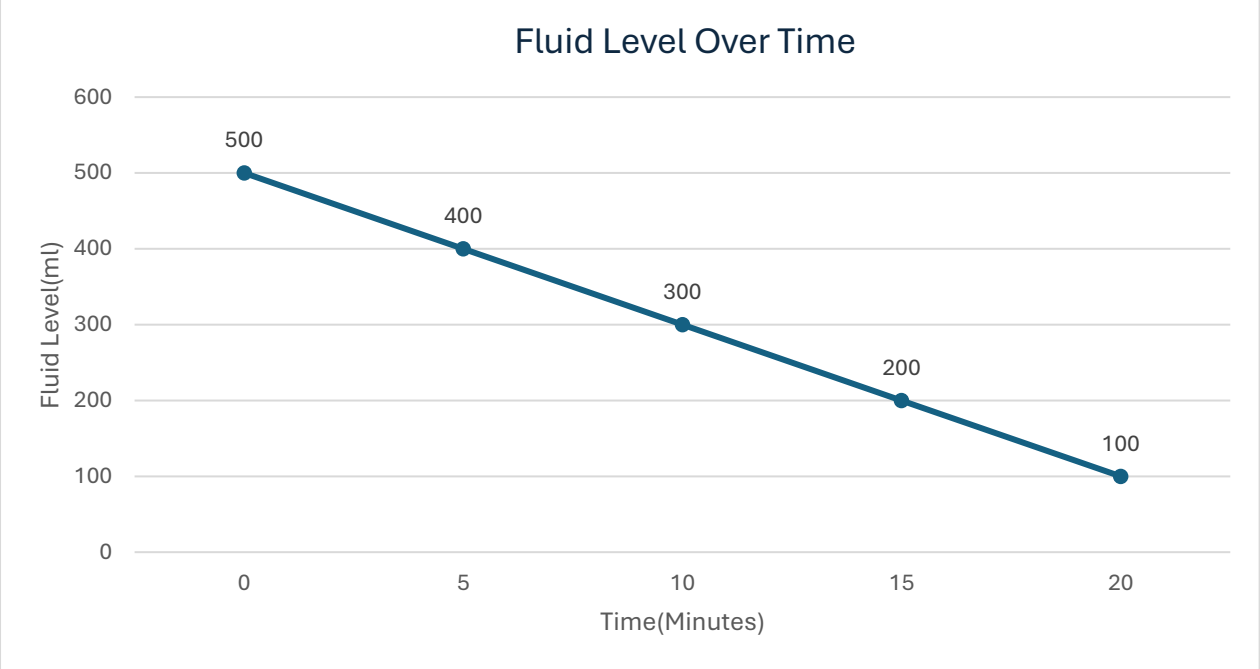


Figure 1 : Illustrate the decrease in IV fluid level over time, with an alert triggered when the threshold is reached

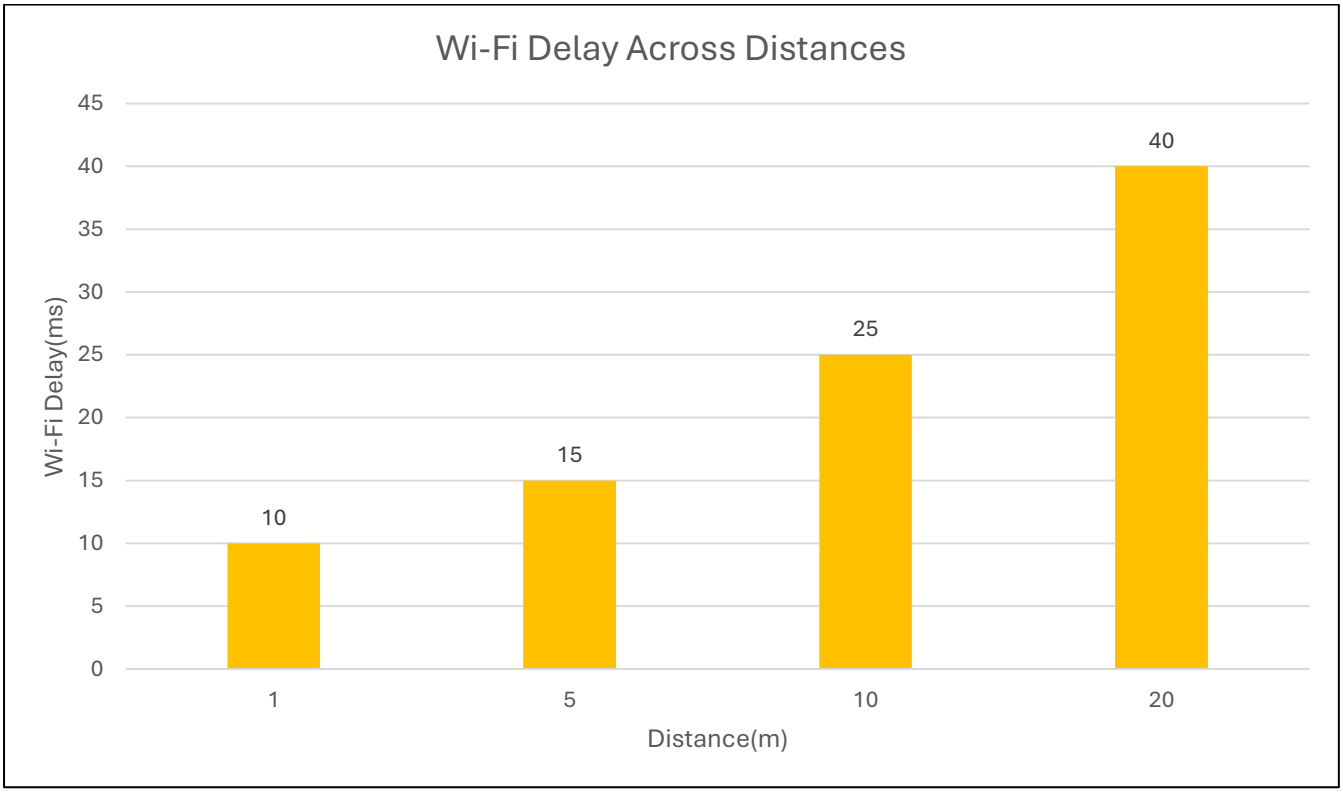


Figure 2: This graph shows the relation of Wi-Fi delay with distance. As the distance increases between the device and the source of Wi-Fi, the delay in communication increases; thus, distance shows its effect on the strength of a signal and the speed at which it is transmitted.

Test Number	Expected Outcome	Measured Outcome	Error (%)
1	Two claps detected	Two claps detected	0%
2	No claps detected	One clap detected	50%
3	Two claps detected	Two claps detected	0%
4	No claps detected	No claps detected	0%

Test Number	Expected Fluid Level(ml)	Measured Fluid Level(ml)	Error (%)
1	500	505	1%
2	1000	980	-2%
3	750	760	1.33%

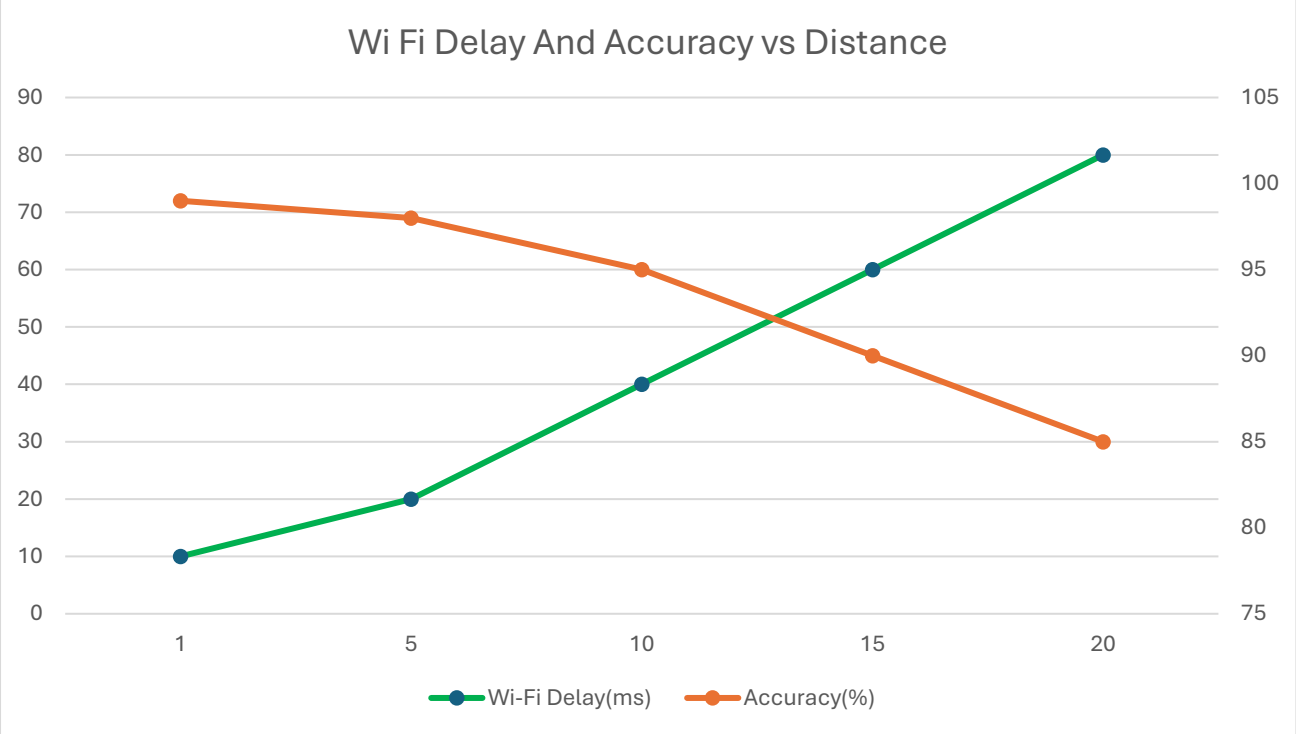


Figure 3: Here is the chart showing Wi-Fi delay and accuracy across different distances. As the distance increases, the Wi-Fi delay rises while the accuracy decreases, highlighting the impact of distance on system performance

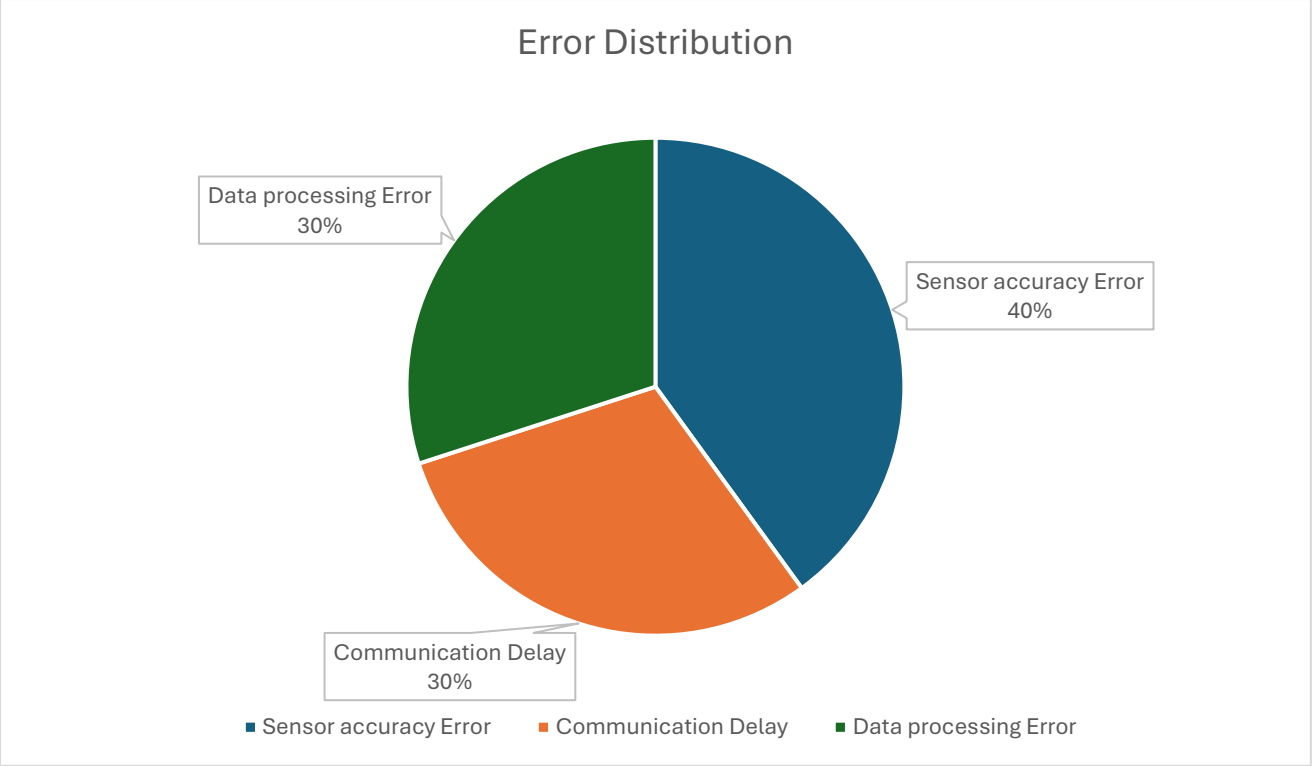


Figure 4: Here is the pie chart showing the distribution of errors in the system. The chart divides errors into categories: ***Sensor Accuracy Error, Communication Delay,*** *and* ***Data Processing Error,*** *with their respective percentages*

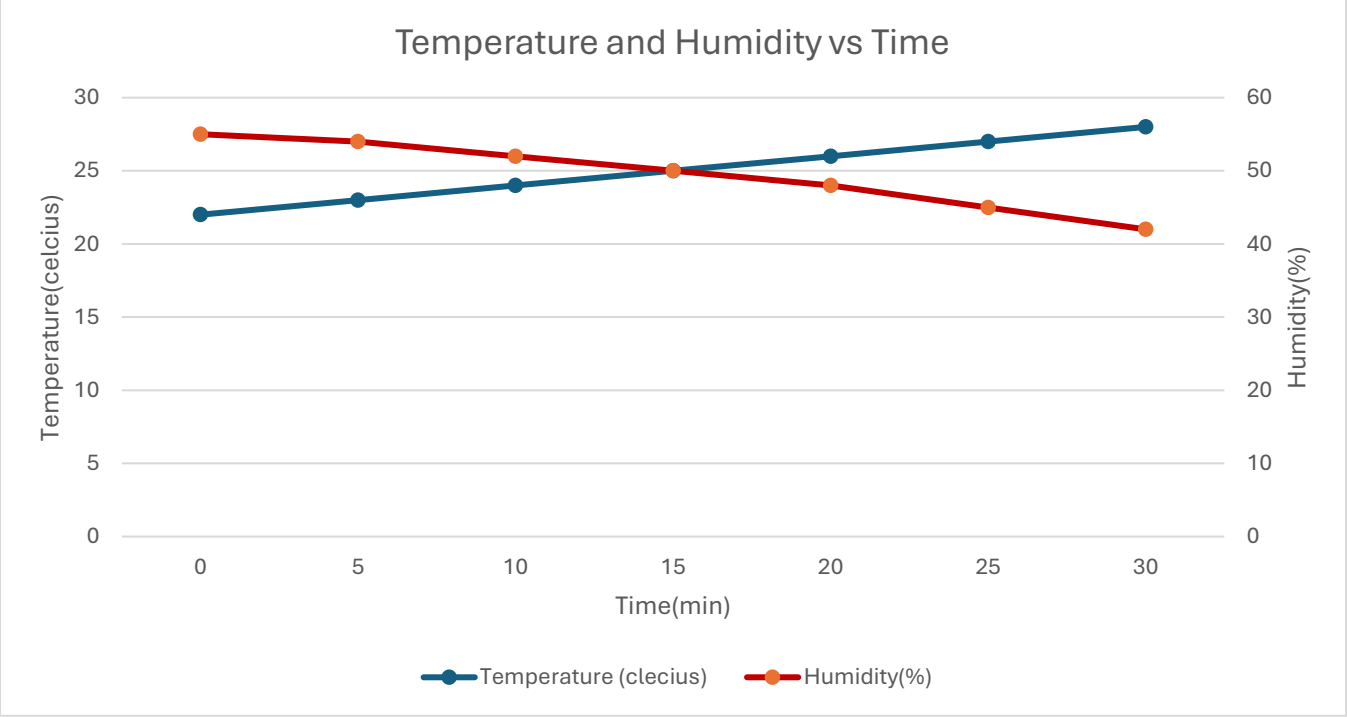


Figure 5: Here is the chart showing Temperature and Humidity vs. Time. The red line represents the temperature, which increases over time, while the blue dashed line represents humidity, which decreases as time progresses. This demonstrates the inverse relationship between temperature and humidity in the recorded data.

Discussion

1.Why did we observe what we observed?

- Accurately calibrated, the weight sensor, and a reliable HX711 module for loa d cell amplification allowed the system to monitor fluid levels efficiently and notify medical staff of low levels. The detection of claps for nurse alerts worked quite effectively in quiet conditions, which showed that the sound sensor was appropriately configured for double claps. On the other hand, degradation in performance occurred at noisy conditions, which seriously affected the detection accuracy.

2.Explanation for Results:

- Expected Results:

The system was true in making the measurement of fluid level accurately because of the exact incorporation of the weight sensor and the microcontroller Arduino Uno. The DHT11 sensor also delivered a continuous reading of temperature and humidity values, hence proving its reliability in medical monitoring applications.

- Unexpected Results:

The interference in Wi-Fi communication caused minor delays in sending notifications. In addition, false detection of claps was recorded in noisy environments, since the sound sensor was triggered by the generally overlapping sound frequencies.

- Hypothesis for Unusual Results:

This could be because the ESP-01 Wi-Fi module has a very limited range and it is meant for short-range applications. The sound sensor may not have been able to differentiate between clapping and the background sounds due to environmental noise. These could be optimized by thresholding of the sensor using advanced sound processing algorithms.

4. Comparison Literature:

- Confirm Previous Studies:
  These findings are consistent with the existing literature showing that fluid monitoring requires calibrated sensors. Similar studies involving sound sensors also note that they face similar problems under conditions of noise, thus designating this as a weakness.

- Deviate from them:
  Unlike most intravenous monitoring systems, our project will integrate a patient self-activated alert mechanism-double clap detection-to call nurses, which has not been widely explored in existing literature. This may offer an added channel of communication in emergency conditions for improved patient care.

Conclusion

- This project demonstrated the development of a feasible IoT-based intravenous fluid monitoring and alerting system, integrating real-time fluid level detection, environmental monitoring, and alerts initiated by the patient for the nurse. The investigation pointed out that the most important issues in healthcare systems are sensor calibration, effective data transmission, and user-friendly alert mechanisms

- Strengths of studies:
  1.Correct monitoring of fluid levels was done using weight sensors and a module named HX711.
  2. Wi-Fi-based notifications for remote have been integrated successfully.
  3. Integrating sound sensors for patient self-activated alerts a truly onovative way to improve patient care

- Weaknesses:
  1. Limited Wi-Fi range sometimes led to delays in receiving notifications.
  2. False positives in clap detection were observed in noisy environments.
  3. Most of these systems depend on stable network connections, which may be a limiting factor in resource-constrained settings.

- Possible applications of the study:
  1. IV fluid monitoring in hospitals and clinics efficiently for great patient care
  2. The home healthcare set-up, particularly for patients with issues in mobility.
  3. Emergency care settings where nurse alerts are very timely.

Recommendations

1.Integrate an Advanced Wi-Fi Module

- Upgrading the Wi-Fi module to an advanced version, like the ESP32, is one of the potential improvements for the system. The ESP32 provides a number of advantages over the ESP-01 module, such as better range, higher data transmission speeds, and dual-core processing capabilities. This would significantly enhance the overall performance of the system, especially in larger medical facilities where signal strength can be inconsistent. This system could support higher data rates with the ESP32 and would provide a stable connection, even in areas with possible interference. The ESP32 also includes built in Bluetooth capabilities, opening up integrations with Bluetooth-enabled devices such as smartphones and wearables that could further enhance remote monitoring capabilities.

2. Improving Noise Immunity by Implementing Advanced Sound Detection Algorithms

- In the most real world applications, There is lot of interface of noise . For improving the accuracy of the system more sophisticated sound detection algorithms should be implemented. Such algorithms may use DSP techniques to filter out the background noise and detect only the intended claps. For example, filtering techniques such as Fourier transforms or wavelet transforms can be applied to isolate the specific frequencies corresponding to the clapping sound. Also, integrating noise-filtering hardware, such as directional microphones or microphone arrays, would allow the system to better distinguish between claps from the ambient sounds even in noisy environments.

3. Test Real-world in Various Settings

- While it should work well under ideal conditions, the real test will indeed be conducted in an on-field environment, with surrounding background noise, interference from signaling, and acoustics from the room, all potential factors to impact performance significantly. The testing of the system under various conditions-for instance, in noisy hospital wards where there is a high-pitched level of background chatter and areas where the Wi-Fi signal is low-will certainly provide valuable inputs into the robustness and reliability of the system. Such

testing will also point to places of failure and indicate improvement that needs to be done so the system can handle the demands placed by real-life conditions. The data collected from such diverse environments will contribute to the fine-tuning of both software and hardware parts of the system to make it function seamlessly in all kinds of scenarios.

4. Introduce a Secondary Alert Mechanism (e.g., Push Buttons)

- While the feature of sound detection is an innovative way for patients to communicate with caregivers, it may be that in very noisy environments or when a patient is too weak to make enough noise, the system may not work as well. In this regard, it would be prudent that a secondary alert mechanism such as push buttons or a call button system should be instituted. These could be installed near the patient's bed for an easy trigger in case they cannot clap or make noise. This would be a fail-safe should the main system malfunction, ensuring the patient is always able to alert the caregiver when in need. Additionally, incorporating several mechanisms of alerts increases the overall reliability of the system.

5. Explore Low-Power Communication Protocols (e.g., Zigbee or LoRa)

- Wi-Fi-based systems may have poor performance when a reliable network is not provided, or the network congestion is high. The solution to these challenges could possibly be the use of communication protocols that are of low power and long-range, like Zigbee or LoRa. These technologies are especially apt for IoT applications, with their longer range and lower power compared to Wi-Fi. Zigbee works in mesh networks, wherein several devices are allowed to communicate over long distances without needing to be connected to the central hub directly. LoRa, on the other hand, can also cover long distances using minimal power and, hence, is ideal for rural or remote areas where access to a reliable Wi-Fi network is poor. These protocols, if implemented, would enhance the system to be more reliable, especially if the Wi-Fi network isn't strong or is intermittent.